

ISHA GUPTA

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EDUCATION

Thapar Institute of Engineering & Technology

Bachelor of Engineering - Computer Science and Engineering

July 2023 – June 2027

CGPA : 8.92

EXPERIENCE

Experiential Learning Centre-TIET

Summer Intern

June 2024 – August 2024

- Programmed and debugged Arduino-based motion control for an **autonomous cleaning robot**, focusing on accurate navigation and turning logic.
- Integrated and configured obstacle detection sensors (e.g., ultrasonic or IR) to support **autonomous path planning and collision avoidance**.
- Utilized **arduino, motor drivers**, and **Sensor modules** to program and fine-tune motion control for an autonomous cleaning robot.

Mehta Family School for Data Science and Artificial Intelligence - IIT Roorkee

Summer Intern

June 2025 – July 2025

- Contributed to research on privacy threats in machine learning systems, with a focus on **Differential Privacy** and its impact on model security.
- Investigated and analyzed privacy attacks such as **membership inference**.
- Simulated and analyzed real-world **privacy attacks** to assess the robustness of **differential privacy** mechanisms.

Samsung PRISM

Research Intern

October 2025 – Feb 2026

- Developing a machine learning model to classify structural, textural, and tonal improvements in **RAW DNG** image buffers.
- Building a Python pipeline for **DNG decoding, RAW buffer extraction**, enhancement, and evaluation.
- Researching image quality metrics and validation methods for enhancement assessment.

TECHNICAL SKILLS

Languages & Programming: Python | C++ | C | R | SQL

Libraries & Frameworks: pandas | numpy | matplotlib | seaborn | scikit-learn | PyTorch | TensorFlow | MXNet | Caffe | Opacus | OpenCV | MediaPipe | YOLO

Web Technologies: HTML | CSS | JavaScript | Node.js | Express.js | React.js | REST API

Databases: MySQL | PostgreSQL | MongoDB

Tools & Platforms: Git | GitHub | Postman | Jupyter Notebook | Google Colab | VS Code | MATLAB

CS Fundamentals: Data Structures and Algorithms | Problem Solving | Competitive Programming

PROJECTS

Hidden Gems | [App Link](#)

- * Users can post about tourist spots or review them and can offer to act as local guide.
- * Supports **authentication** by allowing only logged-in users to add a new gems or edit an existing one.
- * Supports **authorization** by restricting the access to edit gems or delete reviews added by other users.
- * Technologies Used : **HTML, CSS, Nodejs, Expressjs, Passportjs, MongoDB.**

Controlling Game With Hand Gestures

- * Developed an interactive game control system powered by hand gesture recognition using **Python**.
- * Used **MediaPipe** and **OpenCV** to develop gesture recognition logic, trained a custom **YOLO model** on a dataset of three gestures—open palm, finger curl, and closed fist—and integrated it with keyboard controls using **PyAutoGUI** to enable real-time game interaction.

Brain Tumor Classification using Federated Learning

- * Developed a deep learning model to classify brain MRI images into four tumor categories — **Glioma, Meningioma, Pituitary, and No Tumor**.
- * Implemented and compared multiple CNN architectures including **MobileNetV2, DenseNet121, ResNet50, and VGG16** for performance benchmarking.
- * Applied **Federated Learning** by simulating four hospitals (clients) to enable decentralized model training while preserving patient data privacy.
- * Technologies Used : **Python, TensorFlow/Keras, Federated Learning, Deep Learning**.

ACHIEVEMENTS

- **Winner of Makeathon'24: An overnight Hackathon**, organized by **TIET Patiala**.
- Participated and Recognised as **top 10 Projects** Titled: Autonomous Cleaning Robot under the mentorship of Dr.Ashish Singla and Dr.Ravinder Kumar Duvedi during Summer Internship organised by **Experiential Learning Centre'24, TIET**.

POSITION OF RESPONSIBILITIES

- **MLSC**: Conducted workshops on **Competitive Programming** and organized **coding contests** to enhance members' skills and foster a competitive spirit.
- **URJA**: Worked as Executive Committee Head for the college fest.
- **MUDRA (Cultural Society)**: Worked as Executive Committee Member for the society.

PERSONAL SKILLS

- Languages – Fluent in English, Hindi and Punjabi; Beginner in **French** (A1 level).
- Sports – District-level **swimming player** with strong discipline and competitive spirit.